

Vietnamese Company Review Sentiment Analysis

End-to-end pipeline: crawling → preprocessing

→

ML classification → ABSA insights

3,000 reviews from **1900.com.vn**

Vietnamese NLP FastText ABSA

Weak Supervision

3K

REVIEWS

85.6%

ACCURACY

6

MODELS

5

ABSA ASPECTS

01

Problem Statement & Dataset

Why this problem · Data collection · Dataset overview

Why Sentiment Analysis on Reviews?

THE GAP

A 4★ rating may still hide serious complaints about salary or career growth. **Star ratings alone are insufficient** to guide HR decisions.

VIETNAMESE NLP CHALLENGE

Standard tools (VADER, TextBlob) are English-only. Vietnamese has **limited labeled corpora** and requires dedicated word segmentation.

BUSINESS MOTIVATION

Banks & fintech competing for talent need **aspect-level insights** — not aggregated scores — to act on specific pain points.

RESEARCH OBJECTIVES

1. Build a 3-class sentiment classifier for Vietnamese workplace reviews
2. Apply ABSA to identify which aspects drive satisfaction
3. Design a complete pipeline — crawl → label → train → visualize

Dataset — Self-Collected from 1900.com.vn

"We constructed a novel dataset by crawling 10,000 Vietnamese workplace reviews from 1900.com.vn — no pre-existing labeled corpus was available for this domain."

COLLECTION PIPELINE

- ① **Scraper** Traverse 250 listing pages → each company's review pages
- ② **Bloom Filter** Deduplicate already-crawled URLs
- ③ **Kafka Producer / Consumer** Async message queue for resilient ingestion
- ④ **NLP Preprocessing** Normalize, tokenize, remove stopwords
- ⑤ **PostgreSQL** Persistent structured storage via ORM

AIRFLOW SCHEDULE

DAG runs daily at **02:00 AM** — incremental crawl ensures fresh data

DATASET SIZE

3,000

reviews · Vietnamese · Banking /
Fintech

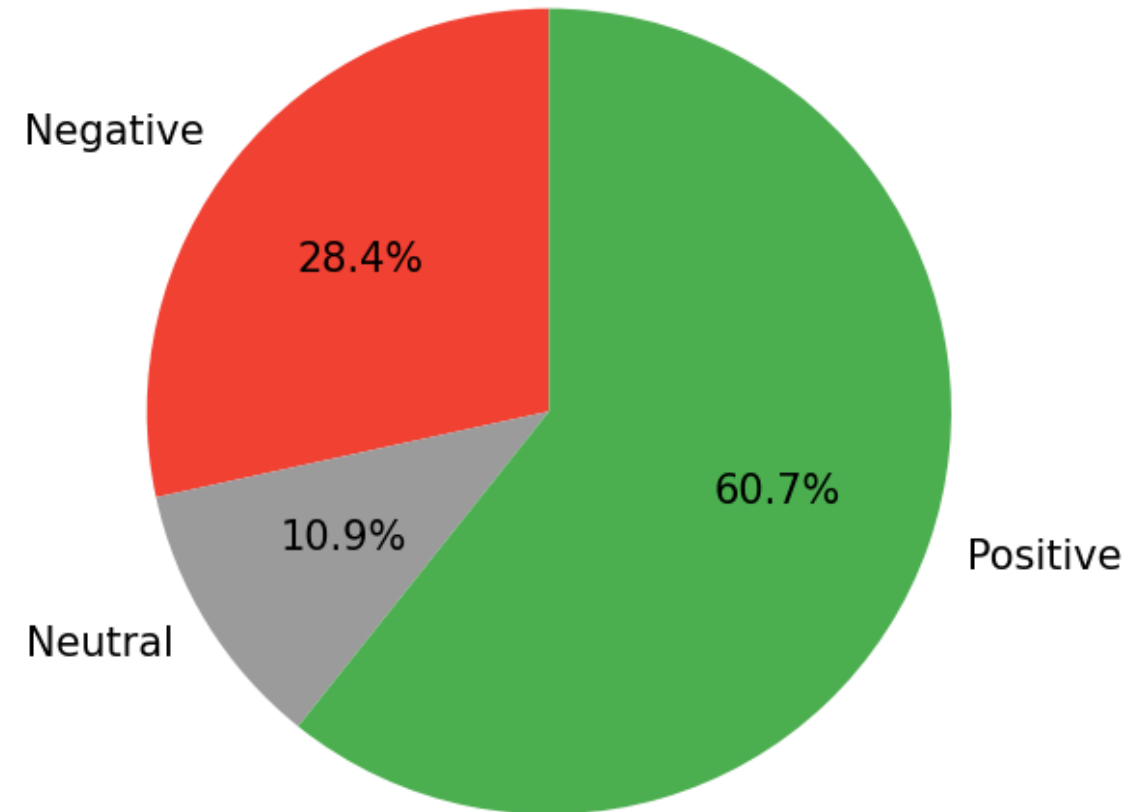
Dataset — Schema & Label Distribution

SCHEMA (10 FIELDS)

Field	Description
rating	1–5 star score
pros	Positive free-text
cons	Negative free-text
advice	Suggestions to management
job_title	Reviewer's role
employee_status	Current / Former
company / industry	Company metadata
location / date	Where & when

CLASS DISTRIBUTION — IMBALANCED

Class Distribution (n=10,000)



02

Preprocessing Pipeline

Vietnamese-specific NLP · 8-step pipeline · underthesea

Text Preprocessing — 8 Steps

`src/preprocessing/processor.py` — Vietnamese text requires dedicated handling different from English

- ① **Unicode NFC Normalization** Standardize Vietnamese diacritics: à á â ã ạ ấ ầ ẩ ố ...
- ② **Lowercase** Reduce vocabulary size, normalize case variants
- ③ **Remove URLs & Emails** Strip `http(s)://...` and `@email` noise
- ④ **Strip HTML Tags** Remove leaked markup from web scraping
- ⑤ **Character Whitelist** Keep Vietnamese chars + digits + `. ! ? ; : -` only
- ⑥ **Collapse Whitespace** Normalize excessive/irregular spacing
- ⑦ **Tokenization — *underthesea*** Vietnamese word segmentation (e.g. "phát_triển", "Hồ_Chí_Minh")
- ⑧ **Stopword Removal** Remove ~60 function words, pronouns, prepositions + English stopwords

Stopword Categories

`src/preprocessing/stopwords_vi.py`

VIETNAMESE FUNCTION WORDS

có, được, là, làm, để, cho, sẽ, đã, đang, bị, phải, nên, cần, muốn...

CONJUNCTIONS & PARTICLES

và, với, nhưng, nếu, khi, mà, vì, hay, hoặc...

PREPOSITIONS

trong, trên, dưới, sau, trước, giữa, qua, đến, vào, ra...

PRONOUNS

tôi, bạn, anh, chị, em, mình, họ, ta, chúng, người, ai...

QUANTIFIERS

các, những, mỗi, mọi, tất, cả, một, nhiều, ít...

ENGLISH STOPWORDS

the, and, is, of, to, in, for, a, an, it, at, be... (mixed-lang reviews)

COMPOUND FRAGMENTS

ty, trường, nghiệp, viên, triển, thiện... (split artifact tokens)

03

Model & Training

Feature engineering · Weak supervision · 6 models · Results

Feature Engineering — 305-dim Input Vector

FASTTEXT EMBEDDINGS — 300 DIMENSIONS

- Pretrained model: `cc.vi.300.bin` (Facebook AI · Vietnamese Common Crawl)
- Weights are **frozen** — used as a semantic feature extractor, not fine-tuned
- Mean pooling over all tokens → one 300-dim vector per review
- Captures semantic similarity across Vietnamese vocabulary

HANDCRAFTED FEATURES — 5 DIMENSIONS

Feature	Description
<code>char_len</code>	Character length (normalized)
<code>word_count</code>	Token count post-tokenization
<code>excl_ratio</code>	Ratio of exclamation marks (!)
<code>pos_ratio</code>	Ratio of positive keywords detected
<code>neg_ratio</code>	Ratio of negative keywords detected

FINAL INPUT VECTOR

305

dimensions

`300` FastText

+

`5` Handcrafted

Labeling Strategy — Weak Supervision

User-assigned star ratings are noisy — a reviewer may give 4★ but write an overwhelmingly negative review. `src/training/labeling.py`

BASE LABELS (FROM STAR RATING)

1-2★ → Negative

3★ → Neutral

4-5★ → Positive

OVERRIDE RULES (SIGNAL-BASED FLIP)

1. **Keyword signal:** Count pos/neg keywords across pros + cons + advice
2. **ABSA signal:** Score aspect opinion terms; `cons` column weighted **1.5×** (complaints are more detailed)
3. **Combined override:**
 - Neutral + signal ≤ -1.5 → flip Negative
 - Neutral + signal $\geq +1.5$ → flip Positive
 - Positive + ABSA ≤ -3 → flip Negative

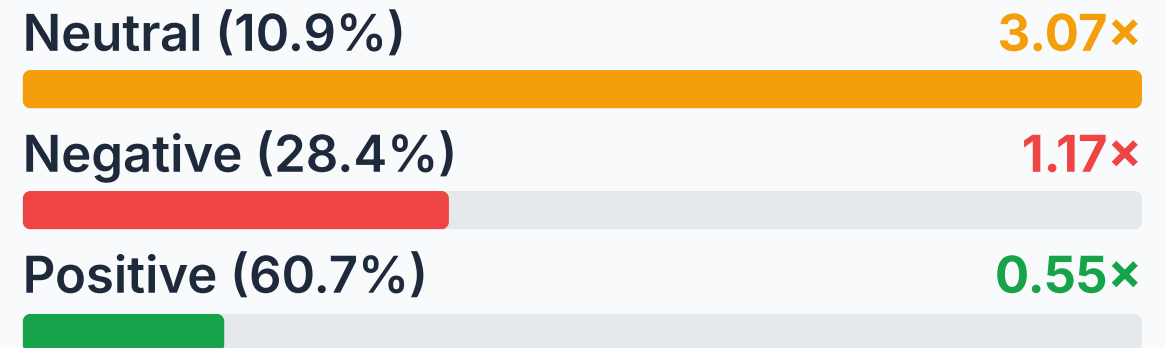
Handling Class Imbalance

METHOD: `class_weight='balanced'`

Applied to all sklearn estimators — adjusts loss function to penalize errors on minority classes more heavily.

Class	Share	Weight
Neutral	10.9%	3.07×
Negative	28.4%	1.17×
Positive	60.7%	0.55×

CLASS WEIGHT VISUALIZATION

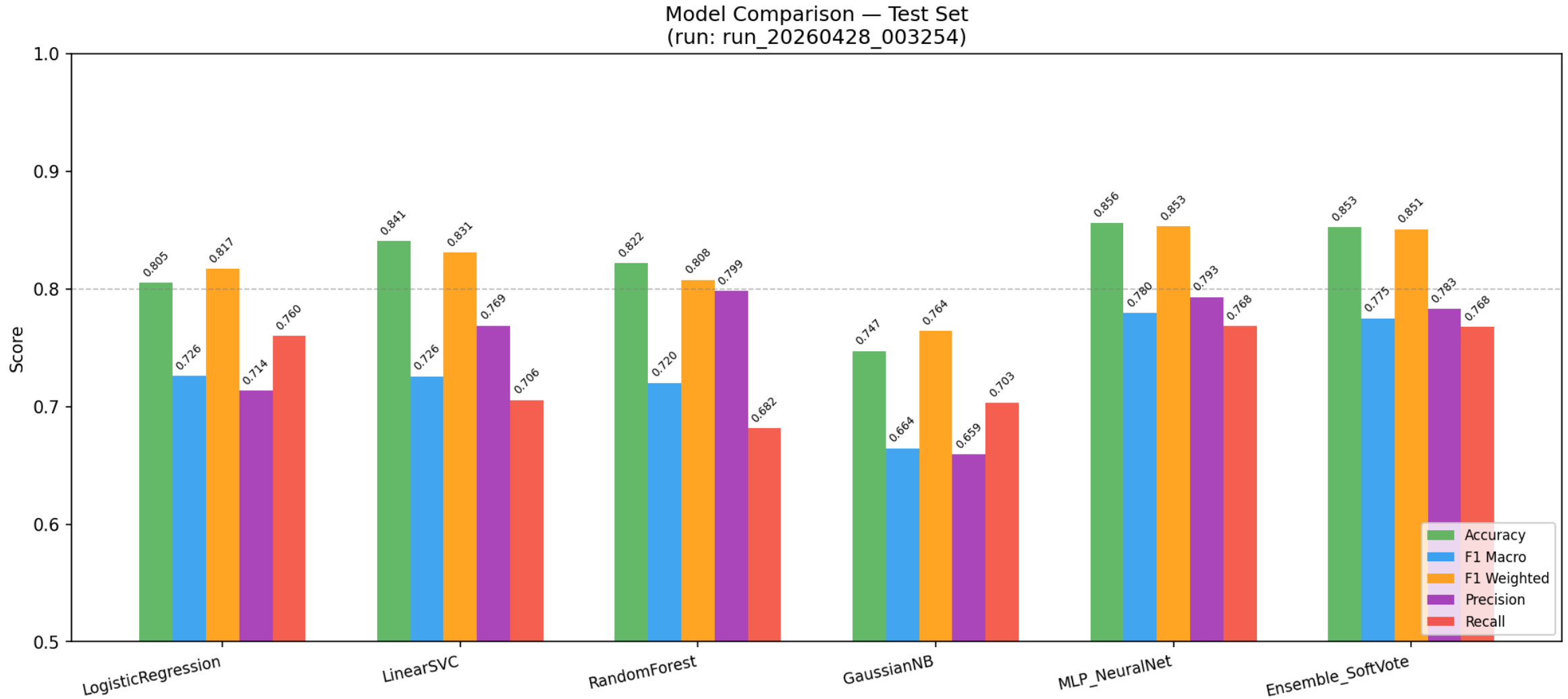


Experiment Results — 6 Models

Split: 70% train / 15% val / 15% test · 10,000 samples total

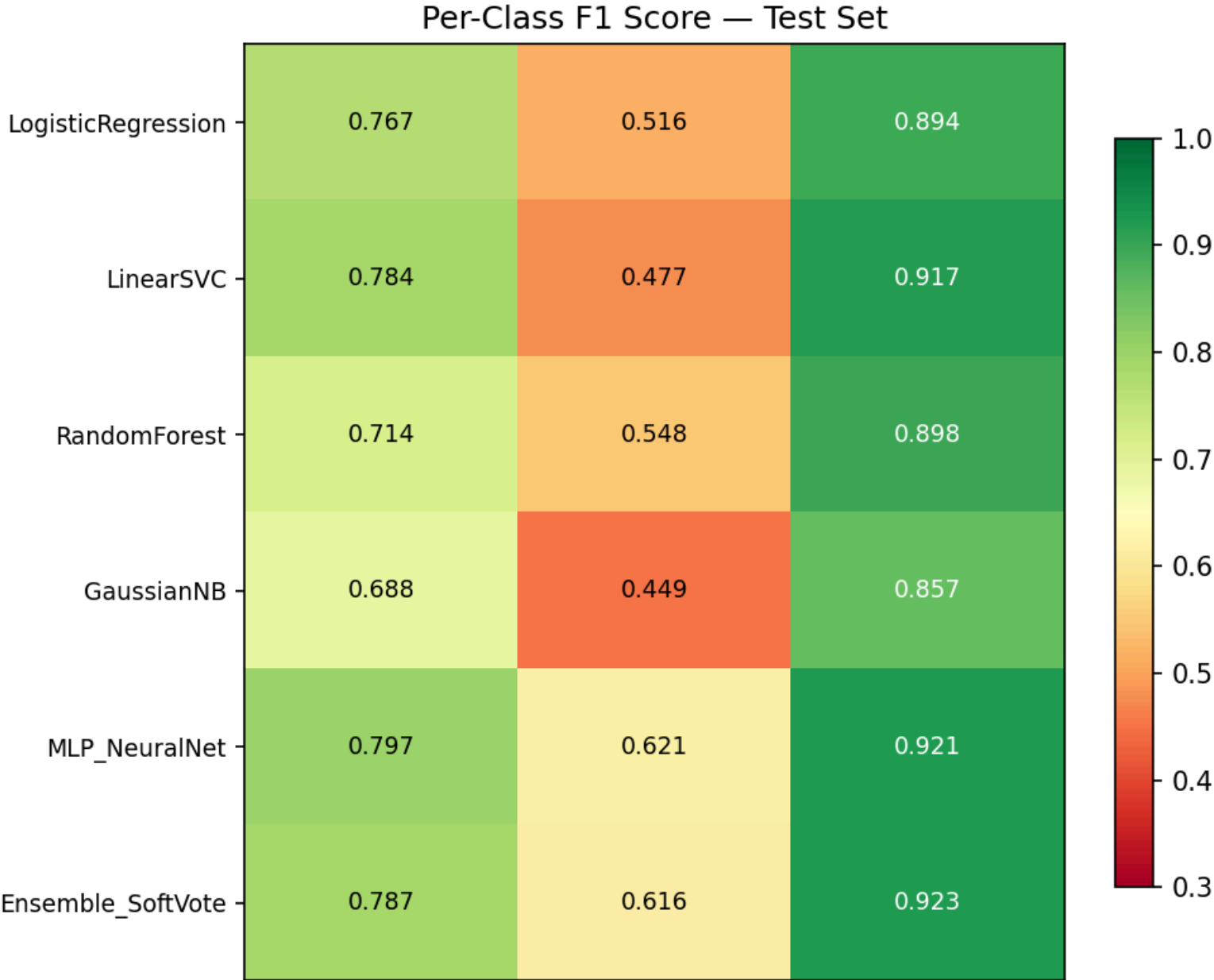
Model	Accuracy	F1 Macro	F1 Weighted	Precision	Recall
★ MLP Neural Net	85.6%	0.780	0.853	0.793	0.769
Ensemble (Soft Vote)	85.3%	0.775	0.851	0.783	0.768
Linear SVC	84.1%	0.726	0.831	0.769	0.706
Random Forest	82.2%	0.720	0.808	0.799	0.682
Logistic Regression	80.5%	0.726	0.817	0.714	0.760
Gaussian NB	74.7%	0.665	0.765	0.659	0.703

Model Comparison — Test Set



MLP_NeuralNet leads across all 5 metrics

Per-Class F1 Score — All Models

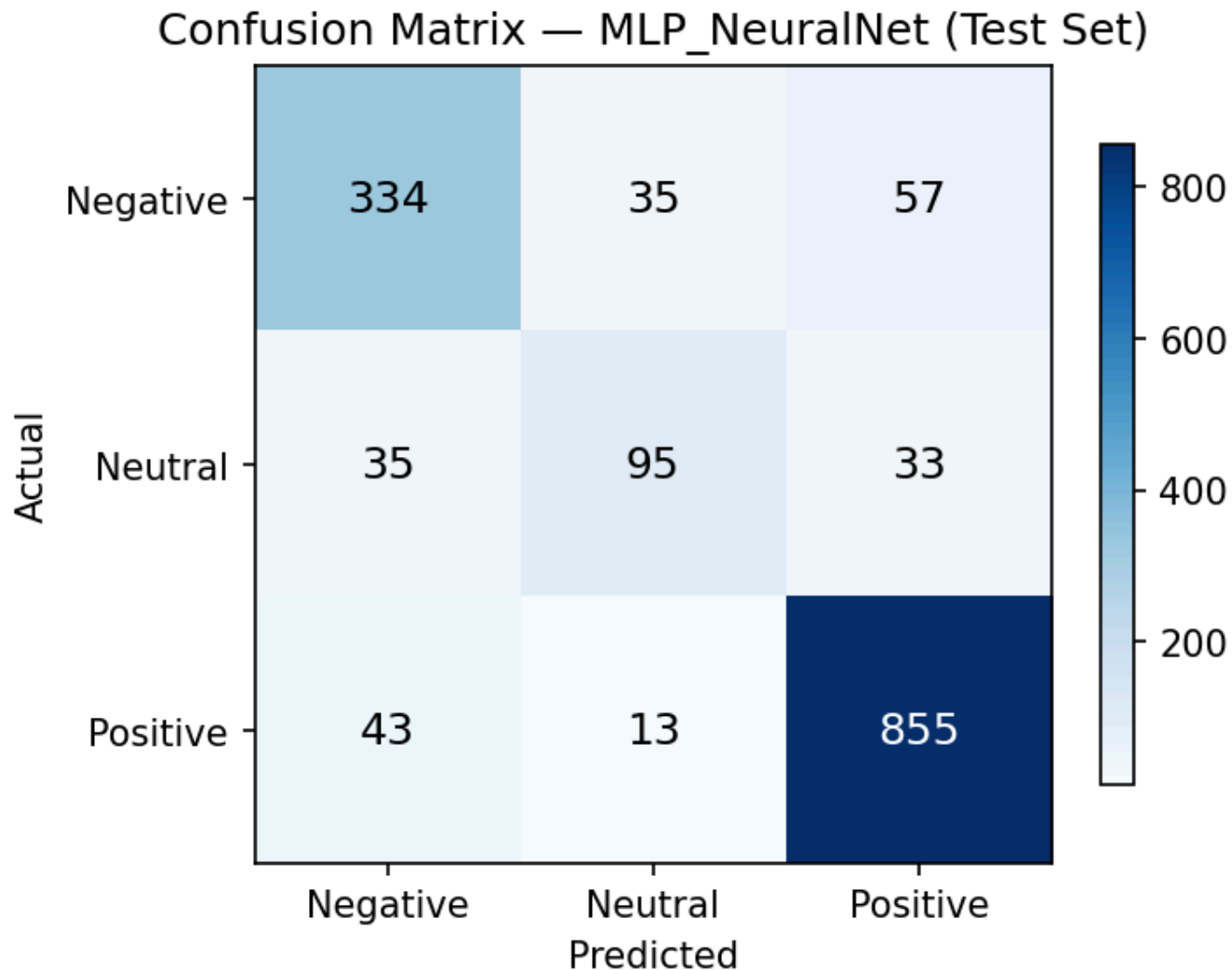


BEST MODEL (MLP)	
Class	F1
Positive	0.921
Negative	0.797
Neutral	0.621

KEY OBSERVATION

Neutral class is the hardest — only **163 test samples** (10.9%). All models struggle here regardless of architecture.

Confusion Matrix — MLP Neural Net



KEY OBSERVATIONS

- **Positive** class: 855/911 correct — 93.8% recall
- **57 negatives** misclassified as positive (13.4% FN rate) — HR risk
- **Neutral**: 68 errors (41.7% error rate) — smallest class, hardest to predict

MLP ARCHITECTURE

256 → 128 → 64 hidden layers
Adam optimizer
Early stopping on val loss

MLP_NeuralNet · Test set (n=1,500)

04

Business Insights

ABSA · 5 workplace aspects · Key findings · HR implications

Aspect Keyword Lexicons

ATE + ACD step — keywords trigger aspect detection; opinion terms extracted from ± 3 token window



Salary & Benefits

lương · thưởng · phúc lợi · thu nhập · đãi ngộ · hoa hồng ·
tăng lương · bảo hiểm · trợ cấp · thù lao · lương thưởng ·
chế độ · benefit



Work Environment

môi trường · văn hóa · đồng nghiệp · văn phòng · không khí
· teamwork · nhóm · tập thể · cởi mở · thân thiện · hòa đồng
· drama · toxic · culture



Management & Leadership

quản lý · lãnh đạo · sếp · giám đốc · trưởng phòng · cấp
trên · ban lãnh đạo · manager · leadership · ban giám đốc ·
tầm nhìn · chính sách



Career Growth

thăng tiến · học hỏi · phát triển · đào tạo · lộ trình · cơ hội ·
kỹ năng · training · kinh nghiệm · thăng chức · học được ·
nâng cao · tiến bộ · career



Work-Life Balance

áp lực · overtime · tăng ca · workload · deadline · cân bằng · giờ làm · stress · quá tải · nghỉ phép · nghỉ ngơi · làm thêm ·
chăm công · giờ giấc

Aspect-Based Sentiment Analysis (ABSA)

Rule-based keyword matching + lexicon opinion extraction · `src/analysis/absa.py`

5 WORKPLACE ASPECTS



Salary & Benefits — lương, thưởng, phúc lợi, đãi ngộ



Work Environment — môi trường, văn hóa, đồng nghiệp



Management — quản lý, lãnh đạo, sếp, giám đốc



Career Growth — thăng tiến, đào tạo, phát triển



Work-Life Balance — áp lực, overtime, deadline, stress

OPINION TERM EXTRACTION (OTE)

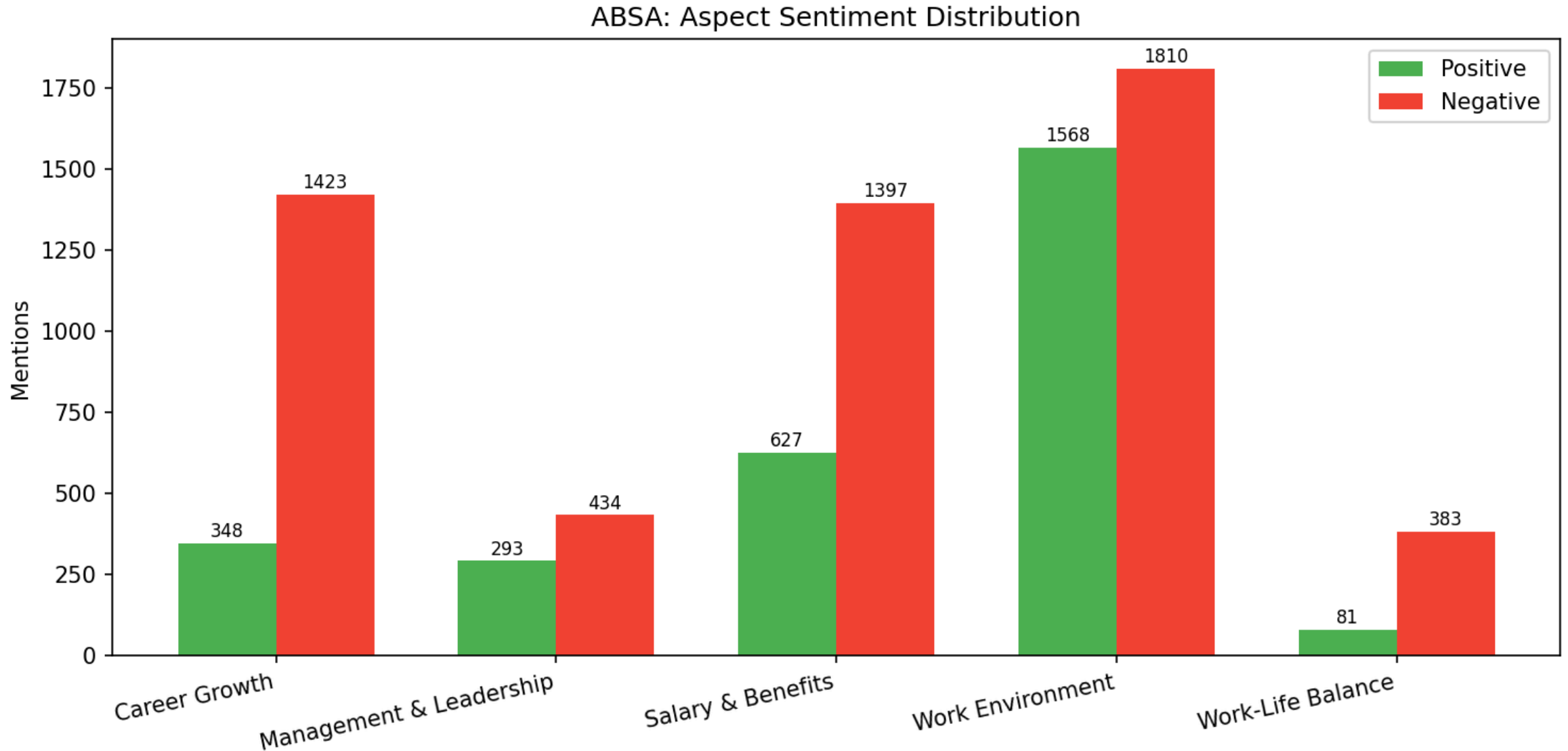
1. Detect aspect keyword in tokenized text
2. Scan ± 3 token window for opinion terms
3. Match against lexicons: 24 positive / 32 negative terms
4. Fallback: infer from source field (pros → positive, cons → negative)

TOTAL MENTIONS

9,354

aspect mentions across 10,000 reviews

ABSA — Aspect Sentiment Distribution



Green – Positive mentions ; Red – Negative mentions ; 5 aspects across 10 000 reviews

Key Findings



Work-Life Balance — 83% negative (81 pos vs 383 neg)

Dominated by: overtime, áp lực, quá tải, stress, deadline. Employees are burning out.



Career Growth — 80% negative (348 pos vs 1,423 neg)

Limited promotion tracks, insufficient training budget. Strongest pain point by volume.



Work Environment — Most discussed (3,378 mentions · 54% negative)

Both the most praised (1,568) and most complained about (1,810). Culture is polarizing — varies by team.



Management — 60% negative (293 pos vs 434 neg)

Keywords: thiếu tầm nhìn, không công bằng, micro-manage. Leadership quality inconsistent.



Salary & Benefits — Most balanced (1,617 pos vs 1,397 neg · 46% neg)

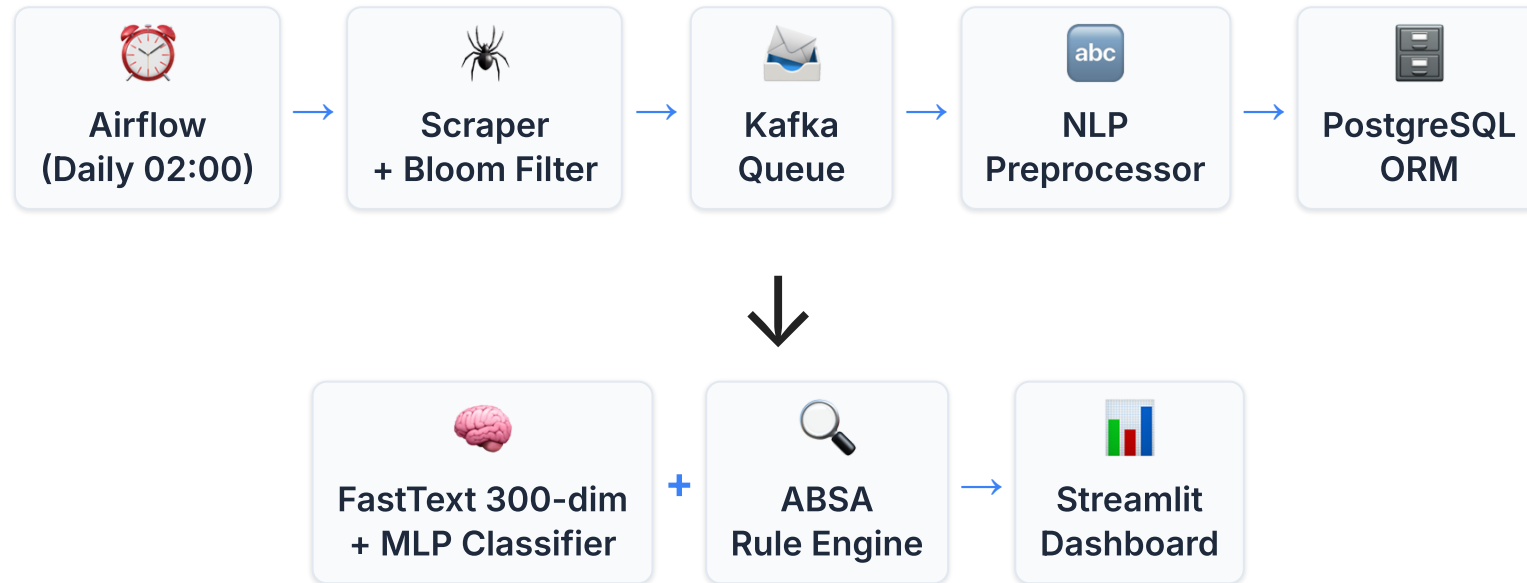
Compensation is competitive but inconsistently applied. Mixed signal — benchmarking recommended.

HR Priority Roadmap

Derived from ABSA negative ratios and mention volumes

Priority	Area	Recommended Action
 High	Work-Life Balance	Review overtime policies; introduce flexible / remote work arrangements
 High	Career Growth	Structured promotion tracks; increase L&D budget; mentorship programs
 Medium	Management	Leadership coaching; implement 360-degree feedback mechanism
 Medium	Work Environment	Culture audit; team-level action plans; address toxic sub-cultures
 Monitor	Salary & Benefits	Compensation benchmarking against fintech market rates; standardize banding

End-to-End System Architecture



Run dashboard: `streamlit run analysis/dashboard.py --server.port 8502`

Summary

WHAT WE BUILT

- Self-collected **10,000-review** Vietnamese dataset
- 8-step NLP preprocessing for Vietnamese text
- Weak supervision labeling from noisy star ratings
- FastText + handcrafted **305-dim** feature vector
- 6 ML models — MLP achieves **85.6%** accuracy
- ABSA over 5 workplace aspects (9,354 mentions)
- Streamlit dashboard for HR analytics

KEY RESULTS

Best accuracy (MLP)	85.6%
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F1 Macro	0.780
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Hardest class (Neutral)	F1 = 0.621
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ABSA mentions	9,354
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Biggest pain point	Work-Life Balance 83% ↓
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